THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE UNIT TEST SET A

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon, and the Sun** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Plants that live on dead matter are called: 1

(a) producers **(b)** parasites **(c)** herbivores **(d)** saprotrophs

2. Plants respire: 1

(a) only during day **(b)** never

(c) all day and night **(d)** only at night

3. The petiole of a leaf is: 1

(a) the root **(b)** the flower

(c) the leaf tip **(d)** the stalk joining leaf to stem

4. Which is a mode of nutrition in which an organism feeds on another living organism? 1

(a) photosynthetic **(b)** saprotrophic **(c)** parasitic **(d)** autotrophic

5. The Sun's light reaches Earth through: 1

(a) water **(b)** vacuum (space) **(c)** glass **(d)** air only

6. A mirror image of 'b' appears as: 1

(a) p upside **(b)** same b

(c) q **(d)** d-like (laterally inverted)

7. The left-right reversal in a mirror is called: 1

(a) refraction **(b)** dispersion

(c) rotation **(d)** lateral inversion

8. The angle of incidence equals the angle of: 1

(a) refraction **(b)** absorption **(c)** reflection **(d)** dispersion

9. The Earth's axis is tilted by about: 1

(a) 90° **(b)** 0° **(c)** 45° **(d)** 23.5°

10. Viewing a solar eclipse directly without protection can damage the: 1

(a) eyes **(b)** lungs **(c)** ears **(d)** skin only

11. Day and night are about equal on the: 1

(a) new moon **(b)** equinoxes **(c)** solstices **(d)** full moon

12. The Earth completes one rotation in about: 1

(a) 365 days **(b)** 30 days **(c)** 12 hours **(d)** 24 hours

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 15–25 words each.

13. How can you test a leaf for starch? 2

14. What three things are needed to form a shadow? 2

15. Why does a shadow of an opaque object have the shape of the object? 2

16. What is revolution? How long does it take? 2

17. What are equinoxes? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. Explain how a pitcher plant catches insects. 3

19. Explain why the image in a pinhole camera becomes blurred when the hole is enlarged. 3

20. Explain how seasons occur due to tilt. 3

21. State safe ways to watch a solar eclipse. 3

22. Explain transport of food in plants. 3

OR

Describe an experiment to show that sunlight is necessary for photosynthesis. 3

23. Differentiate between a shadow and an image in a pinhole camera. 3

OR

An object is 30 cm in front of a plane mirror. Find the distance between the object and its image. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

24. **Case study:** Asha places a ball between a torch and a wall and observes a shadow. 5

(i) Why does it form?
(ii) What if the ball moves closer to the torch?
(iii) What if it is replaced by tracing paper?
(iv) Which property of light is shown?

OR

(i) Define luminous and non-luminous bodies. 5
(ii) Give two examples of each.
(iii) Why is the Moon visible?

25. (i) Define rotation and revolution. 5
(ii) State the time taken by each.
(iii) Name one result of each.

OR

Case study: Neha sees a thin crescent in the evening and a full circle two weeks later. 5

(i) What causes phases?
(ii) Is the Moon waxing or waning first?
(iii) Name the full-moon day.
(iv) Name the new-moon day.

END OF QUESTION PAPER — Total: 50 Marks